

The vertical gradient ($\partial Ri / \partial z$) and curvature ($\partial^2 Ri / \partial z^2$) of the Richardson number are derived using the chain rule on the dimensionless height $\zeta = z/L$.

1. Constant Monin–Obukhov Length ($L = \text{const}$)

When L does not vary with height z , the coordinate transformation gives:

$$\frac{d\zeta}{dz} = \frac{1}{L}, \quad \frac{d^2\zeta}{dz^2} = 0$$

Applying the chain rule yields:

$$\begin{aligned} \frac{\partial Ri}{\partial z} &= \frac{\partial Ri}{\partial \zeta} \frac{d\zeta}{dz} = \frac{1}{L} \frac{\partial Ri}{\partial \zeta} \\ \frac{\partial^2 Ri}{\partial z^2} &= \frac{\partial}{\partial \zeta} \left(\frac{1}{L} \frac{\partial Ri}{\partial \zeta} \right) \frac{d\zeta}{dz} = \frac{1}{L^2} \frac{\partial^2 Ri}{\partial \zeta^2} \end{aligned}$$

- **Equal-Coefficient Case** ($\beta_h = \beta_m = \beta$):

$$\begin{aligned} Ri(\zeta) &= \frac{\zeta}{1 + \beta\zeta} \\ \frac{\partial Ri}{\partial z} &= \frac{1}{L(1 + \beta\zeta)^2} > 0 \\ \frac{\partial^2 Ri}{\partial z^2} &= -\frac{2\beta}{L^2(1 + \beta\zeta)^3} < 0 \end{aligned}$$

This profile exhibits strict negative curvature ($\partial^2 Ri / \partial z^2 < 0$), meaning $Ri(z)$ increases monotonically with height at a decreasing rate, approaching $Ri_c = 1/\beta$ asymptotically.

- **General Case** ($\beta_h \neq \beta_m$):

$$\begin{aligned} Ri(\zeta) &= \frac{\zeta(1 + \beta_h\zeta)}{(1 + \beta_m\zeta)^2} \\ \frac{\partial Ri}{\partial z} &= \frac{1}{L} \frac{1 + (2\beta_h - \beta_m)\zeta}{(1 + \beta_m\zeta)^3} \\ \frac{\partial^2 Ri}{\partial z^2} &= \frac{1}{L^2} \frac{2(\beta_h - 2\beta_m) + 2\beta_m(\beta_m - 2\beta_h)\zeta}{(1 + \beta_m\zeta)^4} \end{aligned}$$

2. Height-Varying Monin–Obukhov Length ($L = L(z)$)

When the surface flux conditions shift with height, $L = L(z)$ and $\zeta(z) = z/L(z)$. The derivatives of ζ with respect to z become:

$$\frac{d\zeta}{dz} = \frac{1 - \zeta L'}{L}, \quad \text{where } L' = \frac{dL}{dz}$$
$$\frac{d^2\zeta}{dz^2} = -\frac{2L'}{L} \left(\frac{1 - \zeta L'}{L} \right) - \frac{\zeta L''}{L} = -\frac{2L'}{L} \frac{d\zeta}{dz} - \frac{\zeta L''}{L}$$

Applying the full total derivative to $Ri(\zeta(z))$ yields:

- **First Derivative (Vertical Gradient):**

$$\frac{dRi}{dz} = \frac{\partial Ri}{\partial \zeta} \left(\frac{1 - \zeta L'}{L} \right)$$

- **Second Derivative (Vertical Curvature):**

$$\frac{d^2 Ri}{dz^2} = \frac{\partial^2 Ri}{\partial \zeta^2} \left(\frac{d\zeta}{dz} \right)^2 + \frac{\partial Ri}{\partial \zeta} \left(\frac{d^2 \zeta}{dz^2} \right)$$
$$\frac{d^2 Ri}{dz^2} = \frac{\partial^2 Ri}{\partial \zeta^2} \left(\frac{1 - \zeta L'}{L} \right)^2 - \frac{\partial Ri}{\partial \zeta} \left[\frac{2L'}{L} \left(\frac{1 - \zeta L'}{L} \right) + \frac{\zeta L''}{L} \right]$$

Key Physical Implications of $L(z)$ Variation:

- **Gradient Modification:** A positive flux gradient ($L' > 0$) reduces $d\zeta/dz$, suppressing the growth of Ri with height compared to the standard MOST profile.
- **Curvature Modulation:** The total vertical curvature $\frac{d^2 Ri}{dz^2}$ splits into two components: the intrinsic MOST stability curvature ($\partial^2 Ri / \partial \zeta^2$) and an explicit boundary-layer curvature induced by flux divergence (L') and shear/heat-flux rate of change (L'').